Dynamic redox-coupling of ceria-based Single-Atom Catalysts

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Abstract

Single-Atom Catalysts (SACs) have emerged as a new frontier in catalysis providing the best noble metal utilization. Atomic dispersion renders the systems highly fluctuous, however, experimental techniques often provide spatial and temporal averages, making full characterization challenging. Reducible oxide supports like ceria further allow for a strong redox coupling with the metal and can facilitate electron-transfer between both[1]. In our work, we discovered the high prevalence of metal oxidation state (mOS) dynamics for a variety of ceria-based SACs, making the common assignment of a fixed oxidation state a too simplistic approximation. We employed data-driven techniques to rationalize charge transfer processes at the metal/oxide interface using fundamental physical descriptors[2]. Lastly, we explored the influence of intrinsic support vacancies and local polaron structure on the dynamic behaviour, showing that surface reduction is a viable route for the stabilization of unusual metal charge states[3].

References

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